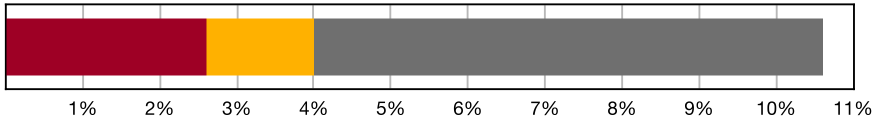
EXTENSION 1
Stage 6

Trigonometry (Ext1)
T1 Inverse Trig Functions (Y11)

Teacher: SHS Maths
Exam Equivalent Time: 91.5 minutes *(based on allocation of 1.5 minutes per mark)*

T1-T3 Trigonometry



- T1 Inverse Trig Functions
- T2 Compound Angles
- T3 Trig Equations

***SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.**

HISTORICAL CONTRIBUTION

- *Trigonometry* has contributed a very healthy 10.6%, on average, to new syllabus Ext1 exams since being introduced in 2020.
- Trigonometry is split into three topic areas for analysis purposes which are: *T1-Inverse Trig Functions (2.6%), T2-Compound Angles (1.4%) and 3-Trig Equations (6.6%).*
- This analysis looks at *T1 Inverse Trig Functions*.

HSC ANALYSIS - What to expect and common pitfalls

- *Inverse Trig Functions (2.6%)* has been examined in two dedicated questions each year between 2022-24.
- "Dedicated question" is used above as this topic area also commonly appears in difficult cross-topic calculus questions that are tested towards the end of the Ext1 exam (not included here).
- Drawing or identifying an inverse trig graph is a common question type, last examined in 2024.
- A deep understanding of domain and range of an inverse function is critical and regularly tested (see *2022 Ext1 13c* and *2024 Ext1 4 MC).*
- Exact values of inverse functions was tested in *2023 Ext1 5 MC* and *2022 Ext1 1 MC*, proving challenging on both occasions.
- Note the syllabus explicitly states that students must understand the notation *arcsin* etc.., which was required in 2023 and 2022 and features in numerous database questions.

Questions

1. Trigonometry, EXT1 T1 2015 HSC 6 MC

What is the domain of the function $f(x) = \sin^{-1}(2x)$?

A. $-\pi \leq x \leq \pi$

B. $-2 \leq x \leq 2$

C. $-\dfrac{\pi}{4} \leq x \leq \dfrac{\pi}{4}$

D. $-\dfrac{1}{2} \leq x \leq \dfrac{1}{2}$

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2. Trigonometry, EXT1 T1 2024 HSC 5 MC

Consider the function $g(x) = 2\sin^{-1}(3x)$.

Which transformations have been applied to $f(x) = \sin^{-1}(x)$ to obtain $g(x)$?

A. Vertical dilation by a factor of $\dfrac{1}{2}$ and a horizontal dilation by a factor of $\dfrac{1}{3}$

B. Vertical dilation by a factor of $\dfrac{1}{2}$ and a horizontal dilation by a factor of 3

C. Vertical dilation by a factor of 2 and a horizontal dilation by a factor of $\dfrac{1}{3}$

D. Vertical dilation by a factor of 2 and a horizontal dilation by a factor of 3

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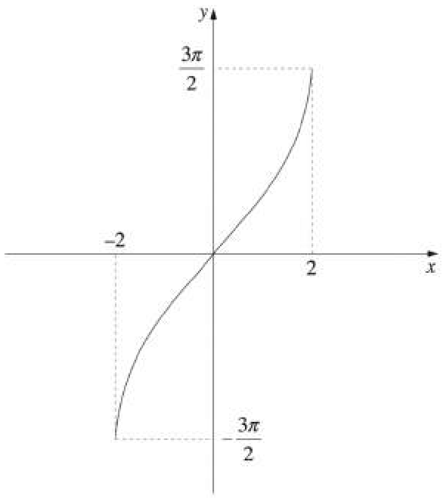
3. Trigonometry, EXT1 T1 2012 HSC 4 MC

Which function best describes the following graph?



A. $y = 3\sin^{-1}2x$

B. $y = \frac{3}{2}\sin^{-1}2x$

C. $y = 3\sin^{-1}\frac{x}{2}$

D. $y = \frac{3}{2}\sin^{-1}\frac{x}{2}$

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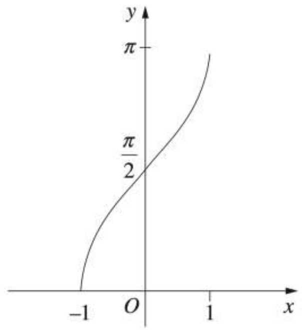
4. Trigonometry, EXT1 T1 2013 HSC 9 MC

The diagram shows the graph of a function.



Which function does the graph represent?

A. $y = \cos^{-1}x$

B. $y = \frac{\pi}{2} + \sin^{-1}x$

C. $y = -\cos^{-1}x$

D. $y = -\frac{\pi}{2} - \sin^{-1}x$

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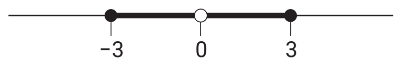
5. Trigonometry, EXT1 T1 2017 HSC 7 MC

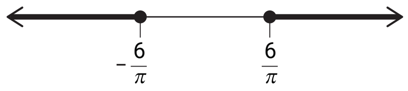
Which diagram represents the domain of the function $f(x) = \sin^{-1}\left(\frac{3}{x}\right)$?

A.

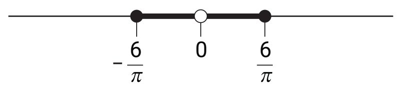
B.



C.



D.



6. Trigonometry, EXT1 T1 2022 HSC 1 MC

It is given that $\cos\left(\frac{23\pi}{12}\right) = \frac{\sqrt{6}+\sqrt{2}}{4}$.

Which of the following is the value of $\cos^{-1}\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right)$?

A. $\frac{23\pi}{12}$

B. $\frac{11\pi}{12}$

C. $\frac{\pi}{12}$

D. $-\frac{11\pi}{12}$

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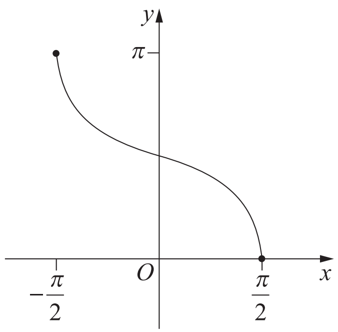
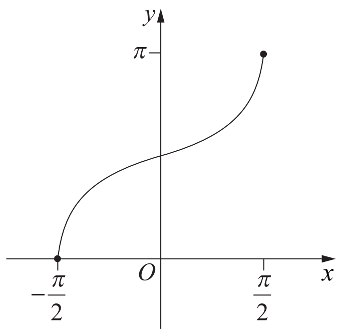
7. Trigonometry, EXT1 T1 2019 HSC 9 MC

Which graph best represents $y = \cos^{-1}(-\sin x)$, for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$?

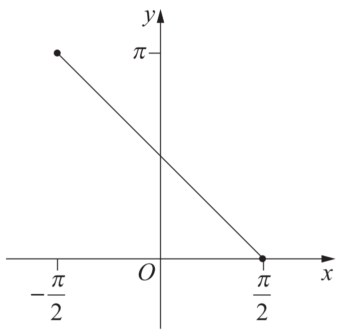
A.



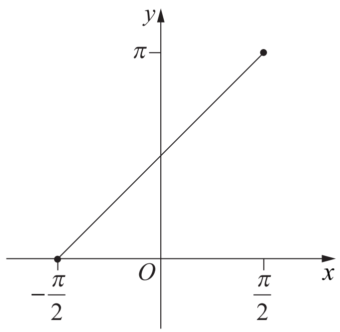
B.



C.



D.



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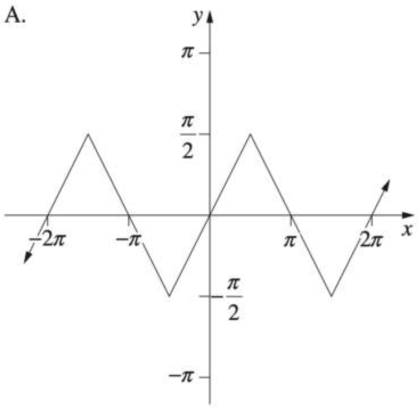
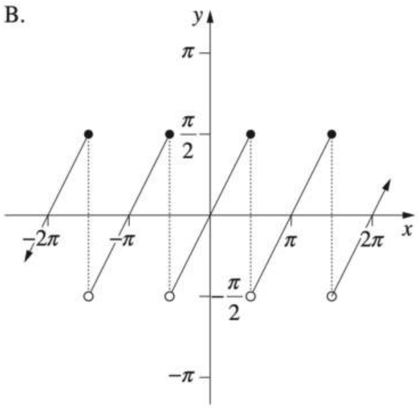
8. Trigonometry, EXT1 T1 2021 HSC 9 MC

Which graph represents the function $y = \sin^{-1}(\sin x)$?

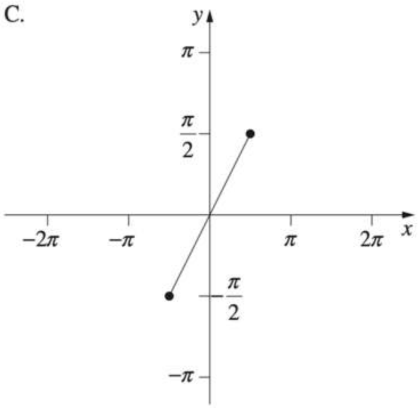
A.



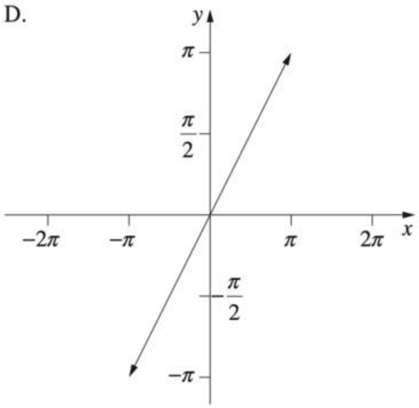
B.



C.



D.



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9. Trigonometry, EXT1 T1 2023 HSC 5 MC

Which of the following is the value of $\sin^{-1}(\sin a)$ given that $\pi < a < \frac{3\pi}{2}$?

A. $a - \pi$

B. $\pi - a$

C. a

D. $-a$

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10. Trigonometry, EXT1 T1 2023 HSC 7 MC

Which statement is always true for real numbers a and b where $-1 \leqslant a < b \leqslant 1$?

A. $\sec a < \sec b$

B. $\sin^{-1} a < \sin^{-1} b$

C. $\arccos a < \arccos b$

D. $\cos^{-1} a + \sin^{-1} a < \cos^{-1} b + \sin^{-1} b$

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11. Trigonometry, EXT1 T1 2024 HSC 4 MC

What are the domain and range of the function $y = 2\cos^{-1}(2x) + 2\sin^{-1}(2x)$?

A. Domain: $[-0.5, 0.5]$ and Range: $\{\pi\}$

B. Domain: $[-0.5, 0.5]$ and Range: $[-\pi, 3\pi]$

C. Domain: $[-2, 2]$ and Range: $\{\pi\}$

D. Domain: $[-2, 2]$ and Range: $[-\pi, 3\pi]$

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12. Trigonometry, EXT1 T1 EQ-Bank 3 MC

The graph of the function $y = \arccos(x-3)$ is transformed by being dilated horizontally with a scale factor of $\dfrac{1}{2}$ and then translated to the left by 1.

What is the equation of the transformed graph?

A. $y = \cos^{-1}\left(\dfrac{x-5}{2}\right)$

B. $y = \cos^{-1}\left(\dfrac{x-8}{2}\right)$

C. $y = \cos^{-1}(2x-1)$

D. $y = \cos^{-1}(2x-2)$

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13. Trigonometry, EXT1 T1 EQ-Bank 4 MC

The graph of the function $y = \sin^{-1}(x-2)$ is transformed by being dilated by a factor of 3 from the y-axis and then translated to the right by 2.

What is the equation of the transformed graph?

A. $y = \sin^{-1}\left(\dfrac{x-8}{3}\right)$

B. $y = \sin^{-1}\left(\dfrac{x-12}{3}\right)$

C. $y = \sin^{-1}(3x-4)$

D. $y = \sin^{-1}(3x-8)$

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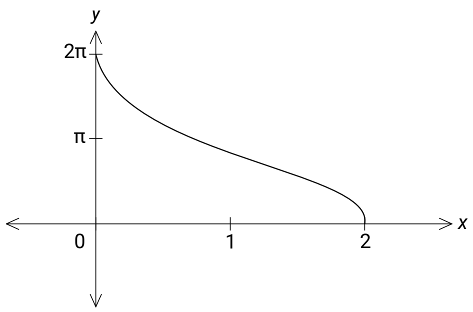
14. Trigonometry, EXT1 T1 EQ-Bank 5 MC

Which equation describes the graph shown?



A. $y = 2\cos^{-1}(x+1)$

B. $y = \sin^{-1}(x-1)$

C. $y = 2\cos^{-1}(x-1)$

D. $y = \dfrac{1}{2}\sin^{-1}(x+1)$

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15. Trigonometry, EXT1 T1 EQ-Bank 2

State the domain and range of the function

$y = \arccos 3x$ ***(2 marks)***

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16. Trigonometry, EXT1 T1 2011 HSC 1e

Find the exact value of $\cos^{-1}\left(-\dfrac{1}{2}\right)$. ***(1 mark)***

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17. Trigonometry, EXT1 T1 2005 HSC 1c

State the domain and range of $y = \cos^{-1}\left(\dfrac{x}{4}\right)$. ***(2 marks)***

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18. Trigonometry, EXT1 T1 2017 HSC 11d

Sketch the graph of the function $y = 2\cos^{-1} x$. ***(2 marks)***

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19. Trigonometry, EXT1 T1 2014 HSC 11c

Sketch the graph $y = 6\tan^{-1} x$, clearly indicating the range. ***(2 marks)***

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20. Trigonometry, EXT1 T1 EQ-Bank 1

Show that $\cos\left(\sin^{-1} q\right) = \sqrt{1-q^2}$ ***(2 marks)***

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21. Trigonometry, EXT1 T1 SM-Bank 2

Evaluate β if $\beta = \cos^{-1}\left(-\dfrac{\sqrt{3}}{2}\right)$. ***(2 marks)***

22. Calculus, EXT1 C2 2006 HSC 2a

Let $f(x) = \sin^{-1}(x+5)$.

i. State the domain and range of the function $f(x)$. ***(2 marks)***

ii. Find the gradient of the graph of $y = f(x)$ at the point where $x = -5$. ***(2 marks)***

iii. Sketch the graph of $y = f(x)$. ***(2 marks)***

23. Trigonometry, EXT1 T1 2007 HSC 2b

Let $f(x) = 2\cos^{-1}x$.

i. Sketch the graph of $y = f(x)$, indicating clearly the coordinates of the endpoints of the graph. ***(2 marks)***

ii. State the range of $f(x)$. ***(1 mark)***

24. Trigonometry, EXT1 T1 EQ-Bank 5

a. State the domain and range for $f(x) = 4\sin^{-1}(2x-3)$. ***(2 marks)***

b. Sketch the function $f(x) = 4\sin^{-1}(2x-3)$. ***(1 mark)***

25. Trigonometry, EXT1 T1 2007 HSC 5c

Find the exact values of x and y which satisfy the simultaneous equations

$\sin^{-1}x + \dfrac{1}{2}\cos^{-1}y = \dfrac{\pi}{3}$ and

$3\sin^{-1}x - \dfrac{1}{2}\cos^{-1}y = \dfrac{2\pi}{3}$. ***(3 marks)***

26. Trigonometry, EXT1 T1 2010 HSC 1b

Let $f(x) = \cos^{-1}\left(\dfrac{x}{2}\right)$. What is the domain of $f(x)$? ***(1 mark)***

27. Trigonometry, EXT1 T1 2011 HSC 2d

Sketch the graph of the function $f(x) = 2\text{arccos}x$. Clearly indicate the domain and range of the function. ***(2 marks)***

28. Trigonometry, EXT1 T1 2012 HSC 13a

Write $\sin\left(2\cos^{-1}\left(\dfrac{2}{3}\right)\right)$ in the form $a\sqrt{b}$, where a and b are rational. ***(2 mark)***

29. Trigonometry, EXT1 T1 EQ-Bank 4

Sketch the graph $y = 2\cos^{-1}(x+1)$. ***(3 marks)***

30. Trigonometry, EXT1 T1 SM-Bank 1

Find $\sin\theta$ given that $\theta = \text{arccos}\left(\dfrac{12}{13}\right) + \text{arctan}\left(\dfrac{3}{4}\right)$, and $0 \le \theta \le \dfrac{\pi}{2}$. ***(3 marks)***

31. Trigonometry, EXT1 T1 SM-Bank 3

Determine the exact value of

$\sin^{-1}\left(\dfrac{\sqrt{3}}{2}\right) - \sin^{-1}\left(-\dfrac{\sqrt{3}}{2}\right) - \cos^{-1}\left(-\dfrac{1}{2}\right)$. ***(3 marks)***

32. Trigonometry, EXT1 T1 SM-Bank 4

Sketch $y = 3\cos^{-1}(2x-1)$ ***(3 marks)***

33. Trigonometry, EXT1 T1 2022 HSC 13c

The function f is defined by $f(x) = \sin(x)$ for all real numbers x. Let g be the function defined on $[-1, 1]$ by $g(x) = \text{arcsin}(x)$.

Is g the inverse of f? Justify your answer. ***(2 marks)***

1. Trigonometry, EXT1 T1 2015 HSC 6 MC

$f(x) = \sin^{-1}(2x)$

Domain of $f(x) = \sin^{-1}x$ is

$-1 \leq x \leq 1$

$\therefore$ Domain of $f(x) = \sin^{-1}(2x)$ is

$-1 \leq 2x \leq 1$

$-\dfrac{1}{2} \leq x \leq \dfrac{1}{2}$

$\Rightarrow D$

2. Trigonometry, EXT1 T1 2024 HSC 5 MC

A vertical dilation of factor 2:

$f(x) = \sin^{-1}(x) \quad \rightarrow \quad f_1(x) = 2\sin^{-1}(x)$

A horizontal dilation of factor $\dfrac{1}{3}$:

$f_1(x) = 2\sin^{-1}(x) \quad \rightarrow \quad f_2(x) = 2\sin^{-1}(3x)$

$\Rightarrow C$

3. Trigonometry, EXT1 T1 2012 HSC 4 MC

Domain: $-2 \leq x \leq 2$

$-1 \leq \dfrac{x}{2} \leq 1$

$\therefore$ Graph is $y = a\sin^{-1}\dfrac{x}{2}$

When $x = 2,\ \ y = \dfrac{3\pi}{2}$:

$\dfrac{3\pi}{2} = a\sin^{-1}1$

$\dfrac{3\pi}{2} = a \times \dfrac{\pi}{2}$

$a = 3$

$\therefore\ y = 3\sin^{-1}\dfrac{x}{2}$

$\Rightarrow C$

4. Trigonometry, EXT1 T1 2013 HSC 9 MC

By elimination,

The graph passes through $(1, \pi)$

The only equation to satisfy this point is

$y = \dfrac{\pi}{2} + \sin^{-1}x$

$\Rightarrow B$

5. Trigonometry, EXT1 T1 2017 HSC 7 MC

$f(x) = \sin^{-1}\left(\dfrac{3}{x}\right)$

$\dfrac{3}{x} \geq -1$ and $\dfrac{3}{x} \leq 1$

$\dfrac{x}{3} \leq -1$ $\qquad x \geq 3$

$x \leq -3$

$\Rightarrow A$

6. Trigonometry, EXT1 T1 2022 HSC 1 MC

Mean mark 54%.

$\dfrac{23\pi}{12} \Rightarrow$ 4th quadrant

$\cos\left(\dfrac{23\pi}{12}\right) = \cos\left(2\pi - \dfrac{23\pi}{12}\right) = \cos\left(\dfrac{\pi}{12}\right)$

$\therefore \cos^{-1}\left(\dfrac{\sqrt{6}+\sqrt{2}}{4}\right) = \dfrac{\pi}{12}$

$\Rightarrow C$

7. Trigonometry, EXT1 T1 2019 HSC 9 MC

$y = \cos^{-1}(-\sin x)$

$= \cos^{-1}\left(\cos\left(x + \dfrac{\pi}{2}\right)\right)$

$= x + \dfrac{\pi}{2}$

$\Rightarrow D$

8. Trigonometry, EXT1 T1 2021 HSC 9 MC

By elimination:

At $x = \pi,\ y = \sin^{-1}(\sin\pi) = \sin^{-1}0 = 0$

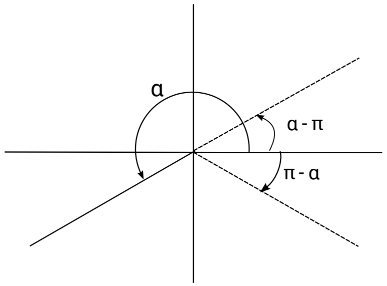
$\rightarrow$ Eliminate C and D

$$\text{At } x = \frac{2\pi}{3},\ y = \sin^{-1}\left(\sin\frac{2\pi}{3}\right) = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$$

$\rightarrow$ Eliminate B

$\Rightarrow A$

9. Trigonometry, EXT1 T1 2023 HSC 5 MC



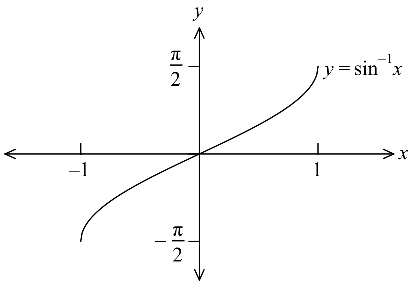
$$\begin{aligned}\sin(a) &= -\sin(a-\pi) \quad \text{(see above)}\\ \sin^{-1}(\sin(a)) &= \sin^{-1}(-\sin(a-\pi))\\ &= \sin^{-1}(\sin(\pi - a))\\ &= \pi - a\end{aligned}$$

Mean mark 56%.

$\Rightarrow B$

10. Trigonometry, EXT1 T1 2023 HSC 7 MC

Consider the graph of $y = \sin^{-1}x$



$y = \sin^{-1}x$ is an increasing function in range $-1 \leqslant x \leqslant 1$

$\therefore$ Given $-1 \leqslant a < b \leqslant 1 \Rightarrow \sin^{-1}a < \sin^{-1}b$

$\Rightarrow B$

11. Trigonometry, EXT1 T1 2024 HSC 4 MC

Domain: $-1 \leqslant 2x \leqslant 1 \Rightarrow -\frac{1}{2} \leqslant x \leqslant \frac{1}{2}$

Range: $2\left(\cos^{-1}(2x) + \sin^{-1}(2x)\right) = 2 \times \frac{\pi}{2} = \pi$

$\Rightarrow A$

12. Trigonometry, EXT1 T1 EQ-Bank 3 MC

$y = \cos^{-1}(x-3)$

Dilate horizontally with scale factor $\frac{1}{2}$

Swap: $x \rightarrow 2x$

$y_1 = \cos^{-1}(2x-3)$

Translate to the left by 1

Swap: $x \rightarrow x+1$

$$\begin{aligned}y_2 &= \cos^{-1}(2(x+1)-3)\\ &= \cos^{-1}(2x+2-3)\\ &= \cos^{-1}(2x-1)\end{aligned}$$

$\Rightarrow C$

13. Trigonometry, EXT1 T1 EQ-Bank 4 MC

$y = \sin^{-1}(x-2)$

Dilate by factor 3 from the y-axis

Swap: $x \to \dfrac{x}{3}$

$$y_1 = \sin^{-1}\left(\frac{x}{3} - 2\right) = \sin^{-1}\left(\frac{x-6}{3}\right)$$

Translate to the right by 2

Swap: $x \to x - 2$

$$y_2 = \sin^{-1}\left(\frac{(x-2)-6}{3}\right) = \sin^{-1}\left(\frac{x-8}{3}\right)$$

$\Rightarrow A$

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14. Trigonometry, EXT1 T1 EQ-Bank 5 MC

Graph shape (general) $\to \cos^{-1}(x)$

Graph translated 1 unit to the right $\to \cos^{-1}(x-1)$

Range: $0 \le \cos^{-1}(x-1) \le \pi \quad \to \quad 0 \le 2\cos^{-1}(x-1) \le 2\pi$

$\Rightarrow C$

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15. Trigonometry, EXT1 T1 EQ-Bank 2

Consider domain: $-1 \leqslant 3x \leqslant 1 \;\Rightarrow\; -\dfrac{1}{3} \leqslant x \leqslant \dfrac{1}{3}$

Domain: $\left[-\dfrac{1}{3}, \dfrac{1}{3}\right]$

Range: $[0, \pi]$

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16. Trigonometry, EXT1 T1 2011 HSC 1e

$\cos\dfrac{\pi}{3} = \dfrac{1}{2}$

Since cos is negative in 2nd quadrant,

$$\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

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17. Trigonometry, EXT1 T1 2005 HSC 1c

$y = \cos^{-1}\dfrac{x}{4}$

Domain of $y = \cos^{-1}x$ is

$-1 \le x \le 1$

$\therefore$ Domain of $y = \cos^{-1}\dfrac{x}{4}$ is

$$-1 \le \frac{x}{4} \le 1$$
$$-4 \le x \le 4$$

Range $y = \cos^{-1} x$ is

$0 \le y \le \pi$

$\therefore$ Range $y = \cos^{-1}\dfrac{x}{4}$ is

$0 \le y \le \pi$

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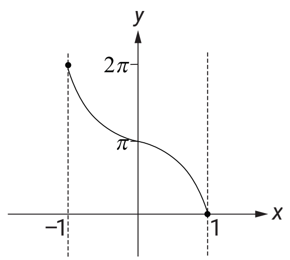
18. Trigonometry, EXT1 T1 2017 HSC 11d

$y = 2\cos^{-1}x$

$\Rightarrow$ Domain $-1 \le x \le 1$

Range $y = 2\cos^{-1}x$ is $0 \le y \le \pi$

$\Rightarrow$ Range $y = 2\cos^{-1}x$ is $0 \le y \le 2\pi$



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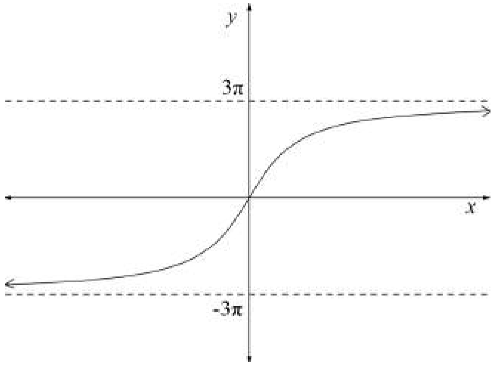
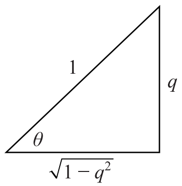
19. Trigonometry, EXT1 T1 2014 HSC 11c

$y = 6\tan^{-1}x$



20. Trigonometry, EXT1 T1 EQ-Bank 1



$\sin\theta = q$

$\theta = \sin^{-1}q$

$\therefore \cos\left(\sin^{-1}q\right) = \cos\theta$

$= \sqrt{1-q^2}$

21. Trigonometry, EXT1 T1 SM-Bank 2

Base angle: $\cos\dfrac{\pi}{6} = \dfrac{\sqrt{3}}{2}$

$\cos \Rightarrow$ $-$ve in 2nd quadrant

$\cos^{-1}\left(-\dfrac{\sqrt{3}}{2}\right) = \pi - \dfrac{\pi}{6}$

$= \dfrac{5\pi}{6}$

22. Calculus, EXT1 C2 2006 HSC 2a

i. $f(x) = \sin^{-1}(x+5)$

Domain

$-1 \le x+5 \le 1$

$-6 \le x \le -4$

Range

$-\dfrac{\pi}{2} \le y \le \dfrac{\pi}{2}$

ii. $y = \sin^{-1}(x+5)$

$\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-(x+5)^2}}$

When $x = -5$

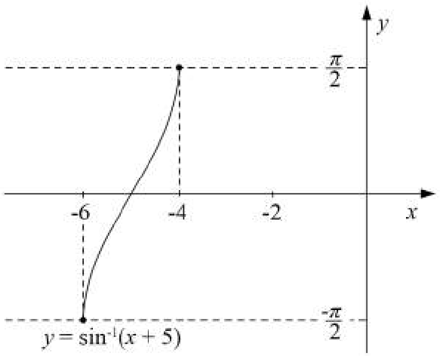
$\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-(-5+5)^2}}$

$= \dfrac{1}{\sqrt{1-0}}$

$= 1$

$\therefore$ Gradient of $y = f(x)$ at $x = -5$ is 1.

iii.



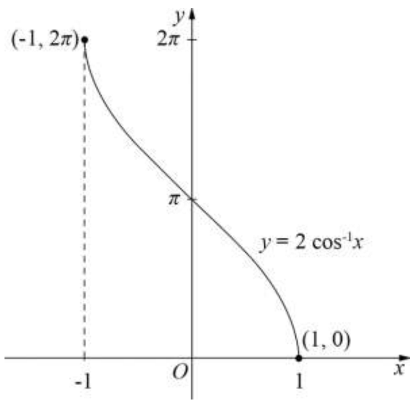
23. Trigonometry, EXT1 T1 2007 HSC 2b

i. $y = 2\cos^{-1}x$

Domain: $-1 \le x \le 1$

Range: $0 \le \dfrac{y}{2} \le \pi$

$0 \le y \le 2\pi$



ii. Range: $0 \le y \le 2\pi$

24. Trigonometry, EXT1 T1 EQ-Bank 5

a. Domain: $-1 \leqslant 2x - 3 \leqslant 1$

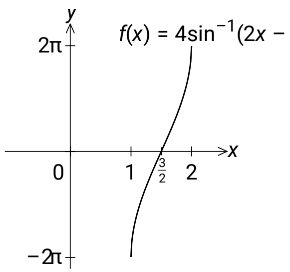
$2 \leqslant 2x \leqslant 4$

$1 \leqslant x \leqslant 2$

Range: $-\dfrac{\pi}{2} \leqslant y \leqslant \dfrac{\pi}{2}$

$-2\pi \leqslant 4y \leqslant 2\pi$

b.



25. Trigonometry, EXT1 T1 2007 HSC 5c

$$\sin^{-1}x + \frac{1}{2}\cos^{-1}y = \frac{\pi}{3} \quad \ldots (1)$$

$$3\sin^{-1}x - \frac{1}{2}\cos^{-1}y = \frac{2\pi}{3} \quad \ldots (2)$$

Add $(1) + (2)$

$$4\sin^{-1}x = \pi$$

$$\sin^{-1}x = \frac{\pi}{4}$$

$$\therefore x = \frac{1}{\sqrt{2}}$$

MARKER'S COMMENT: The use of ther elimination method here proved much more successful than substitution.

Substitute $x = \dfrac{1}{\sqrt{2}}$ into (1)

$$\sin^{-1}\frac{1}{\sqrt{2}} + \frac{1}{2}\cos^{-1}y = \frac{\pi}{3}$$

$$\frac{\pi}{4} + \frac{1}{2}\cos^{-1}y = \frac{\pi}{3}$$

$$\frac{1}{2}\cos^{-1}y = \frac{\pi}{12}$$

$$\cos^{-1}y = \frac{\pi}{6}$$

$$\therefore y = \frac{\sqrt{3}}{2}$$

26. Trigonometry, EXT1 T1 2010 HSC 1b

$$f(x) = \cos^{-1}\left(\frac{x}{2}\right)$$

$$-1 \le \frac{x}{2} \le 1$$

$$-2 \le x \le 2$$

$\therefore$ Domain of $f(x)$ is $-2 \le x \le 2$

27. Trigonometry, EXT1 T1 2011 HSC 2d

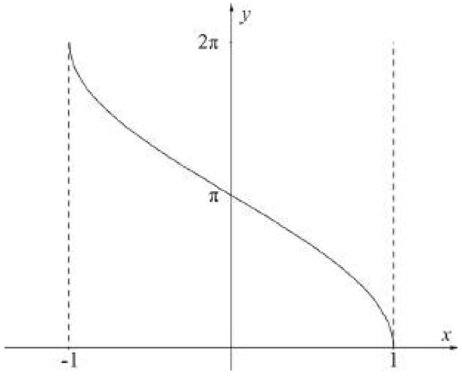
$y = 2\cos^{-1}x$

$\dfrac{y}{2} = \cos^{-1}x$

Domain $-1 \leq x \leq 1$

Since $0 \leq \dfrac{y}{2} \leq \pi$

Range $0 \leq y \leq 2\pi$



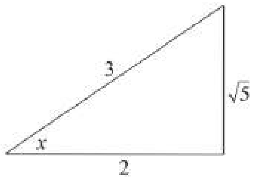
28. Trigonometry, EXT1 T1 2012 HSC 13a

TIP: This question is made less complicated by reminding yourself that $\cos^{-1}\left(\frac{2}{3}\right)$ is simply an angle.

$\sin\left(2\cos^{-1}\left(\dfrac{2}{3}\right)\right)$

Let $x = \cos^{-1}\left(\dfrac{2}{3}\right)$

$\therefore\ \cos x = \dfrac{2}{3}$

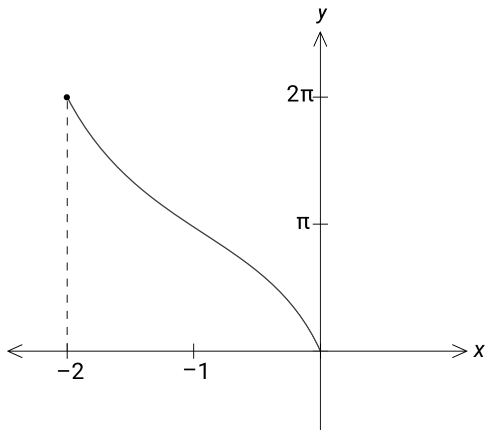


$\sin x = \dfrac{\sqrt{5}}{3}$

$$\begin{aligned}\sin\left(2\cos^{-1}\left(\frac{2}{3}\right)\right) &= \sin 2x \\ &= 2\sin x\cos x \\ &= 2 \times \frac{\sqrt{5}}{3} \times \frac{2}{3} \\ &= \frac{4}{9}\sqrt{5}\end{aligned}$$

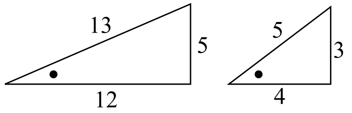
29. Trigonometry, EXT1 T1 EQ-Bank 4

Domain:
$-1 \leqslant x+1 \leqslant 1 \ \Rightarrow \ -2 \leqslant x \leqslant 0$
Range:
$0 \leqslant \cos^{-1}(x) \leqslant \pi \ \Rightarrow \ 0 \leqslant 2\cos^{-1}(x) \leqslant 2\pi$



30. Trigonometry, EXT1 T1 SM-Bank 1

$$\begin{aligned}\sin\theta &= \sin\left(\cos^{-1}\left(\frac{12}{13}\right)+\tan^{-1}\left(\frac{3}{4}\right)\right)\\ &= \sin\left(\cos^{-1}\left(\frac{12}{13}\right)\right)\cos\left(\tan^{-1}\left(\frac{3}{4}\right)\right)+\sin\left(\tan^{-1}\left(\frac{3}{4}\right)\right)\cos\left(\cos^{-1}\left(\frac{12}{13}\right)\right)\end{aligned}$$



$$\begin{aligned}\therefore \sin\theta &= \frac{5}{13}\times\frac{4}{5}+\frac{3}{5}\times\frac{12}{13}\\ &= \frac{20}{65}+\frac{36}{65}\\ &= \frac{56}{65}\end{aligned}$$

COMMENT: Syllabus states that students must understand and use the notation $\arccos$, $\arcsin$ etc..

31. Trigonometry, EXT1 T1 SM-Bank 3

Base angle: $\sin\dfrac{\pi}{3}=\dfrac{\sqrt{3}}{2}$

$\sin \Rightarrow$ – ve in 4th quadrant

$$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=\frac{\pi}{3}$$

$$\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)=-\frac{\pi}{3}$$

Base angle: $\cos\dfrac{\pi}{3}=\dfrac{1}{2}$

$\cos \Rightarrow$ – ve in 2nd quadrant

$$\cos^{-1}\left(-\frac{1}{2}\right)=\pi-\frac{\pi}{3}=\frac{2\pi}{3}$$

$$\begin{aligned}\therefore \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)-\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)-\cos^{-1}\left(-\frac{1}{2}\right) &= \frac{\pi}{3}-\left(-\frac{\pi}{3}\right)-\frac{2\pi}{3}\\ &= 0\end{aligned}$$

32. Trigonometry, EXT1 T1 SM-Bank 4

Domain:

$-1 \le (2x - 1) \le 1$

$0 \le 2x \le 2$

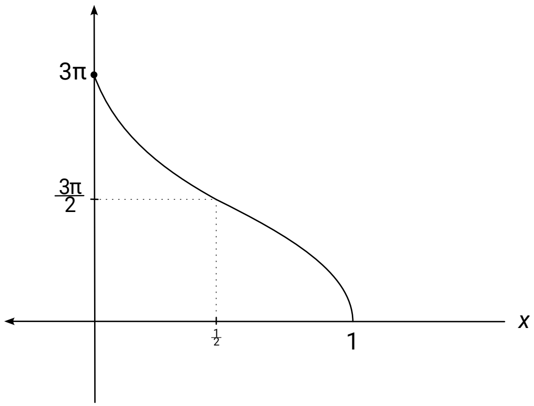
$0 \le x \le 1$

Range of $\cos^{-1}(x) = [0, \pi]$

$\Rightarrow$ Range of $3\cos^{-1}(x) = [0, 3\pi]$

At $x = 0,\ \ y = 3\cos^{-1}(-1) = 3\pi$

Sketch $y = 3\cos^{-1}(2x - 1)$:



33. Trigonometry, EXT1 T1 2022 HSC 13c

The domain of $f(x) = \mathbb{R}$

The range of $g(x) = \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$

Since the domain $f(x) \neq$ range of $g(x)$,

$f^{-1}(x) \neq g(x)$

♦♦ Mean mark 24%.